ICT 1201

Fundamentals of Computer Systems

Number Systems

Introduction

- A set of values used to represent different quantities is known as “Number System”.
- E.g. A number system can be used to represent number of students in a class or numbers of viewers watching a certain TV program etc.
- The digital computer represent all kinds of data and information including text, audio, graphics, video etc. in binary numbers.

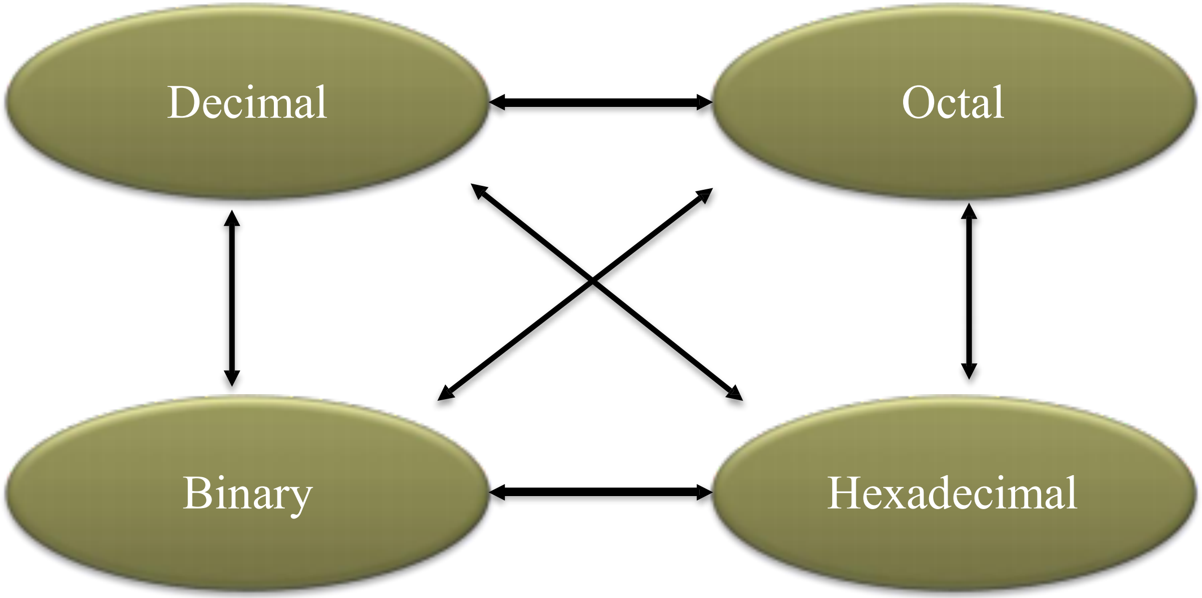
Common Number Systems

System	Base	Symbol	Used by Humans?	Used in Computers?
Decimal	10	0,1,2,3,4,5,6,7,8,9	Yes	No
Binary	2	0,1	No	Yes
Octal	8	0,1,2,3,4,5,6,7	No	No
Hexadecimal	16	0,1,2,3,4,5,6,7,8,9, A,B,C,D,E,F	No	No

- Number of distinct symbols used in a number system is known as its Radix or Base.
  - E.g. In decimal number system the base would be 10 as we use 10 distinct symbols from 0-9

Conversion Among Bases

- The Possibilities



Quick Example

$$25_{10} = 11001_2 = 31_8 = 19_{16}$$

Base / Radix

Decimal to Decimal

$$
\begin{array}{lcccccccccc}
125_{10} & = & 5 & \times & 10^0 & = & 5 & \times & 1 & = & 5 \\
 & & 2 & \times & 10^1 & = & 2 & \times & 10 & = & 20 \\
 & & 1 & \times & 10^2 & = & 1 & \times & 100 & = & 100 \\
\hline
 & & & & & & & & & & 125
\end{array}
$$

Base / Radix

weight

Binary to Decimal

Technique

- Multiply each bit by 2^n, where *n* is the "weight" of the bit.
- The weight is the position of the bit, starting from 0 on the right.
- Add the results.

Example 01

$$101011_2 \Rightarrow \begin{array}{rcrcrcr} 1 \times 2^0 & = & 1 \times 1 & = & 1 \\ 1 \times 2^1 & = & 1 \times 2 & = & 2 \\ 0 \times 2^2 & = & 0 \times 4 & = & 0 \\ 1 \times 2^3 & = & 1 \times 8 & = & 8 \\ 0 \times 2^4 & = & 0 \times 16 & = & 0 \\ 1 \times 2^5 & = & 1 \times 32 & = & 32 \\ \hline & & & & 43_{10} \end{array}$$

Example 02

$$
\begin{array}{lllllllll}
1101.01_2 = & 1 & \times & 2^{-2} & = & 1 & \times & 0.25 & = 0.25 \\
 & 0 & \times & 2^{-1} & = & 0 & \times & 0.5 & = 0 \\
 & 1 & \times & 2^{0} & = & 1 & \times & 1 & = 1 \\
 & 0 & \times & 2^{1} & = & 0 & \times & 2 & = 0 \\
 & 1 & \times & 2^{2} & = & 1 & \times & 4 & = 4 \\
 & 1 & \times & 2^{3} & = & 1 & \times & 8 & = \underline{8\qquad} \\
 & & & & & & & & \ 13.25_{10}
\end{array}
$$

i. $10001011_2 = ?_{10}$

ii. $1101.101_2 = ?_{10}$

Octal to Decimal

Technique

- Multiply each bit by 8^n, where n is the "weight" of the bit.
- The weight is the position of the bit, starting from 0 on the right.
- Add the results.

Example 01

$$724_8 \Rightarrow \begin{array}{cccccccccc} 4 & x & 8^0 & = & 4 & x & 1 & = & 4 \\ 2 & x & 8^1 & = & 2 & x & 8 & = & 16 \\ 7 & x & 8^2 & = & 7 & x & 64 & = & 448 \\ \hline & & & & & & & & 468_{10} \end{array}$$

Example 02

$$127.4_8 \Rightarrow \begin{array}{llllllllll} 4 & x & 8^{-1} & = & 4 & x & 0.125 & = & 0.5 \\ 7 & x & 8^{0} & = & 7 & x & 1 & = & 7 \\ 2 & x & 8^{1} & = & 2 & x & 8 & = & 16 \\ 1 & x & 8^{2} & = & 1 & x & 64 & = & 64 \\ \hline & & & & & & & & 87.5_{10} \end{array}$$

i. $647_8 = ?_{10}$

ii. $21.21_8 = ?_{10}$

Hexadecimal to Decimal

Technique

- Multiply each bit by 16^n, where n is the "weight" of the bit.
- The weight is the position of the bit, starting from 0 on the right.
- Add the results.

Example 01

$$
2ABC_{16} = \begin{array}{cccccccc}
C & x & 16^0 & = & 12 & x & 1 & = & 12 \\
B & x & 16^1 & = & 11 & x & 16 & = & 176 \\
A & x & 16^2 & = & 10 & x & 256 & = & 2560 \\
2 & x & 16^3 & = & 2 & x & 4096 & = & 8192 \\
\hline
 & & & & & & & & 10940_{10}
\end{array}
$$

i. $15F_{16} = ?_{10}$

ii. $B65F_{16} = ?_{10}$

Decimal to Binary

Technique

- To convert a decimal integer into binary, keep dividing by 2 until the quotient is 0. Keep track of remainder while dividing and collect the remainders in reverse order.
- To convert a fraction, keep multiplying the fractional part by 2 until it becomes 0. Collect the integer in forward order.

Example 01

$125_{10} \Rightarrow ?_2$

2	125	
2	62	1
2	31	0
2	15	1
2	7	1
2	3	1
2	1	1
	0	1

$125_{10} = 1111101_2$

Example 02

$162.375_{10} \Rightarrow ?_2$

2	162	
2	81	0
2	40	1
2	20	0
2	10	0
2	5	0
2	2	1
2	1	0
	0	1

$0.375 \times 2 = 0.750$

$0.750 \times 2 = 1.500$

$0.500 \times 2 = 1.000$

$162.375_{10} = 10100010.011_2$

i. $244_{10} = ?_2$

ii. $254.25_{10} = ?_2$

Octal to Binary

Technique

- Replace each octal digit with its equivalent 3 bit binary sequence

Octal	Binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Example 01

$$705_8 \;\Rightarrow\; ?_2$$

$$\begin{array}{ccc} 7 & 0 & 5 \\ \downarrow & \downarrow & \downarrow \\ 111 & 000 & 101 \end{array}$$

$$705_8 = 111000101_2$$

Example 02

$$673.12_8 \Rightarrow ?_2$$

$$\begin{array}{cccccc} 6 & 7 & 3 & . & 1 & 2 \\ \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow \\ 110 & 111 & 011 & . & 001 & 010 \end{array}$$

$$673.12_8 = 110111011.001010_2$$

i. $632_8 = ?_2$

ii. $512.35_8 = ?_2$

Hexadecimal to Binary

Technique

- Replace each hex digit with its equivalent 4 bit binary sequence.

Hex	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111

Hex	Binary
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

Example 01

$$10AF_{16} \Rightarrow ?_2$$

$$\begin{array}{cccc} 1 & 0 & A & F \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 0001 & 0000 & 1010 & 1111 \end{array}$$

$$10AF_{16} = 0001000010101111_2$$

Example 02

$$261.35_{16} \Rightarrow ?_2$$

$$\begin{array}{cccccc} 2 & 6 & 1 & . & 3 & 5 \\ \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow \\ 0010 & 0110 & 0001 & . & 0011 & 0101 \end{array}$$

$$261.35_{16} = 001001100001.00110101_2$$

i. $AF_{16} = ?_2$
ii. $F27.42_{16} = ?_2$

Decimal to Octal

Technique

- To convert a decimal integer into octal, keep dividing by 8 until the quotient is 0. Keep track of remainder while dividing and collect the remainders in reverse order.
- To convert a fraction, keep multiplying the fractional part by 8 until it becomes 0. Collect the integer in forward order.

Example 01

$1234_{10} \Rightarrow ?_8$

8	1234	
8	154	2
8	19	2
8	2	3
	0	2

$1234_{10} = 2322_8$

Example 02

$$134.75_{10} \Rightarrow ?_8$$

8	134	
8	16	6
8	2	0
	0	2

$$0.75 \quad \times \quad 8 \quad = \quad 6.000$$

$$134.75_{10} = 206.6_8$$

i. $512_{10} = ?_8$

ii. $467.125_{10} = ?_8$

Decimal to Hexadecimal

Technique

- To convert a decimal integer into ~~octal,~~ Hexadecimal keep dividing by 16 until the quotient is 0. Keep track of remainder while dividing and collect the remainders in reverse order.
- To convert a fraction, keep multiplying the fractional part by 16 until it becomes 0. Collect the integer in forward order.

Example 01

$1234_{10} \Rightarrow ?_{16}$

$$\begin{array}{r|ll}
16 & 1234 & \\
16 & 77 & 2 \\
\hline
16 & 4 & 13=D \\
\hline
 & 0 & 4
\end{array}$$

$1234_{10} = 4D2_{16}$

i. $5324_{10} = ?_{16}$
ii. $2130_{10} = ?_{16}$

Binary to Octal

Technique

- Make groups of 3 bits, starting from the binary point.
- Add 0 s to the end of the number (to the fraction part) if needed.
- Convert each bit group to its corresponding octal digit.

Octal	Binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Example 01

$1011010111_2 \quad => ?_8$

$$\begin{array}{cccc} 001 & 011 & 010 & 111 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 3 & 2 & 7 \end{array}$$

$1011010111_2 \quad = 1327_8$

Example 02

$$10110100.001011_2 \Rightarrow ?_8$$

$$\begin{array}{cccccc} 010 & 110 & 100 & . & 001 & 011 \\ \downarrow & \downarrow & \downarrow & & \downarrow & \downarrow \\ 2 & 6 & 4 & . & 1 & 3 \end{array}$$

$$10110100.001011_2 = 264.13_8$$

i. $110100010011_2 = ?_8$

ii. $101100.0011011_2 = ?_8$

Binary to Hexadecimal

Technique

- Make groups of 4 bits, starting from the binary point.
- Add 0 s to the end of the number (to the fraction part) if needed.
- Convert each bit group to its corresponding hexadecimal digit.

Hex	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111

Hex	Binary
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

Example 01

$1010111011_2 \Rightarrow ?_{16}$

$$\begin{array}{ccc} 0010 & 1011 & 1011 \\ \downarrow & \downarrow & \downarrow \\ 2 & B & B \end{array}$$

$1010111011_2 = 2BB_{16}$

i. $1101110010011_2 = ?_{16}$

ii. $101100.0011011_2 = ?_{16}$

Octal to Hexadecimal

Technique

- Use binary as an intermediary.

Octal	Binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Example 01

$1076_8 \;\; => ?_{16}$

1 0 7 6

↓ ↓ ↓ ↓

001 | 000 11 | 1 110

2 | 3 | E

$1076_8 \;\; = 23E_{16}$

i. $1156_8 = ?_{16}$

ii. $325.23_8 = ?_{16}$

Hexadecimal to Octal

Technique

- Use binary as an intermediary

Hex	Binary
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111

Hex	Binary
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

Example 01

$1F0C_{16} \Rightarrow ?_8$

1	F	0	C
↓	↓	↓	↓
0001	1111	0000	1100

$0|001\ |111|1\ \ 000|0\ \ 1|100$

1 | 7 | 4 | 1 | 4

$1F0C_{16} = 17414_8$

i. $A4B6_{16} = ?_8$

ii. $2E75.C2_{16} = ?_8$

Exercise – Fill the blanks

Decimal	Binary	Octal	Hexadecimal
33			
	1110101		
		703	
			1AF

Answer

Decimal	Binary	Octal	Hexadecimal
33	100001	41	21
117	**1110101**	165	75
451	111000011	**703**	1C3
431	110101111	657	**1AF**

Summary

Conversion	Operation
Decimal to Binary	Decimal number divide by 2
Decimal to Octal	Decimal number divide by 8
Decimal to Hexadecimal	Decimal number divide by 16
Binary to Decimal	Each digit multiply by 2
Octal to Decimal	Each digit multiply by 8
Hexadecimal to Decimal	Each digit multiply by 16
Binary to Octal	Divide digits into 3 groups and write equivalent octal value
Binary to Hexadecimal	Divide digits into 4 groups and write equivalent hex value
Octal to Binary	Write equivalent binary values for each octal digit
Hexadecimal to Binary	Write equivalent binary values for each hex digit
Octal to Hexadecimal	Use binary as an intermediary
Hexadecimal to Octal	Use binary as an intermediary

THANK YOU